

LAB REPORT: DYNAMIC NAT VS. PAT CONFIGURATION

Cisco CCNA Lab Implementation & Analysis — Cisco Packet Tracer

Router: R1 (Cisco ISR)

Topic: NAT Pool Exhaustion & Port Address Translation

Status: Successfully Completed

1. EXECUTIVE OVERVIEW & TOPOLOGY

This lab report demonstrates the step-by-step configuration, verification, error handling, and operational differences between **Dynamic NAT (Address Pooling)** and **PAT (Port Address Translation / NAT Overload)** on Cisco IOS routers.

Device / Interface	IP Address / Subnet	NAT Role	Description
R1 - GigabitEthernet0/1	172.16.0.254 /24	IP NAT INSIDE	Internal LAN Gateway (SW1, PC1, PC2, PC3)
R1 - GigabitEthernet0/0	203.0.113.1 /30	IP NAT OUTSIDE	Public WAN Interface to Internet ISP
Internal Subnet	172.16.0.0 /24	Inside Local	Hosts: PC1 (.1), PC2 (.2), PC3 (.3)
Dynamic NAT Pool	100.0.0.1 – 100.0.0.2	Inside Global Pool	Allocated pool size: 2 Public IP Addresses
Internet DNS Server	8.8.8.8	Outside Global	External Destination Target

2. TASK 1: DYNAMIC NAT CONFIGURATION & SYNTAX TROUBLESHOOTING

Goal: Configure Dynamic NAT translating internal hosts (172.16.0.0/24) using a public pool of two IPs (100.0.0.1 to 100.0.0.2).

CLI Troubleshooting Note (Cisco IOS Syntax Enforcement)

During the pool creation, several syntax variations were attempted. Cisco IOS requires the netmask keyword or exact prefix length notation depending on the software release:

- ip nat pool P00L1 100.0.0.1 100.0.0.2 24 → **ERROR** Invalid input at '^' marker (bare number rejected).
- ip nat pool P00L1 100.0.0.1 100.0.0.2 /24 → **ERROR** Invalid input at '^' marker (slash notation rejected).
- ip nat pool P00L1 100.0.0.1 100.0.0.2 255.255.255.0 → **ERROR** Missing netmask keyword.
- ip nat pool P00L1 100.0.0.1 100.0.0.2 netmask 255.255.255.0 → **SUCCESS** Correct syntax accepted.

```
! Step 1: Define Inside and Outside Interfaces
R1# configure terminal
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip nat outside
R1(config-if)# exit
R1(config)# interface GigabitEthernet0/1
R1(config-if)# ip nat inside
R1(config-if)# exit

! Step 2: Define Standard Access List matching internal LAN
R1(config)# access-list 1 permit 172.16.0.0 0.0.0.255

! Step 3: Define NAT Pool (with correct netmask keyword)
R1(config)# ip nat pool P00L1 100.0.0.1 100.0.0.2 netmask 255.255.255.0
```

```
! Step 4: Map ACL to Dynamic NAT Pool
R1(config)# ip nat inside source list 1 pool POOL1
```

3. TASK 2: DYNAMIC NAT TESTING & POOL EXHAUSTION ANALYSIS

Test Scenario: Initiate ping requests to google.com / 8.8.8.8 sequentially from PC1, PC2, and PC3.

Host Name	Source IP	Target	Ping Result	Assigned Inside Global IP	Status / Observation
PC1	172.16.0.1	172.217.175.238	4/4 SUCCESS (0% LOSS)	100.0.0.1	Successfully translated using 1st pool IP.
PC2	172.16.0.2	8.8.8.8	4/4 SUCCESS (0% LOSS)	100.0.0.2	Successfully translated using 2nd pool IP.
PC3	172.16.0.3	8.8.8.8	3 TIMED OUT, 1 RECEIVED	NONE (Pool Exhausted)	Failed initially due to 1-to-1 IP pool depletion!

Technical Explanation of PC3 Ping Failure:

Dynamic NAT creates 1-to-1 temporary bindings without port overloading. Since **POOL1** contains only 2 public IP addresses (100.0.0.1 and 100.0.0.2), PC1 and PC2 immediately consumed all available public addresses. When PC3 initiated traffic, R1 dropped packets until an entry in the NAT translation table timed out. The single successful response occurred only after a translation entry expired.

4. TASK 3 & 4: TRANSITION TO PAT (PORT ADDRESS TRANSLATION)

To eliminate address starvation, the dynamic NAT pool was removed and replaced with **PAT (NAT Overload)** utilizing R1's public interface IP address (GigabitEthernet0/0 - 203.0.113.1).

```
! Step 1: Clear existing translation table and remove dynamic pool
R1# clear ip nat translation *
R1(config)# no ip nat pool POOL2 100.0.0.0 100.0.0.1 netmask 255.255.255.0

! Step 2: Configure PAT using Interface Overload with ACL 2
R1(config)# access-list 2 permit 172.16.0.0 0.0.0.255
R1(config)# ip nat inside source list 2 interface GigabitEthernet0/0 overload
```

PAT Verification Results Across All Hosts:

Ping Test Summary (PAT Mode)

- PC1: Pinging google.com → 0% Loss
- PC2: Pinging google.com → 0% Loss
- PC3: Pinging google.com → 0% Loss

NAT Statistics Output Highlights

- **Total Translations:** 3 (0 static, 3 dynamic, 3 extended)
- **Hits:** 27 | **Misses:** 28
- **Inside Interface:** GigabitEthernet0/1
- **Outside Interface:** GigabitEthernet0/0

Active Translation Table Output (show ip nat translations):

```
R1# show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
```

```
udp 203.0.113.1:1024 172.16.0.3:1025 8.8.8.8:53 8.8.8.8:53
udp 203.0.113.1:1025 172.16.0.2:1025 8.8.8.8:53 8.8.8.8:53
udp 203.0.113.1:1026 172.16.0.1:1026 8.8.8.8:53 8.8.8.8:53
```

5. DYNAMIC NAT VS. PAT KEY TAKEAWAYS

Feature	Dynamic NAT (Pool)	PAT (Overload)
Mapping Type	1-to-1 Mapping (IP to IP)	Many-to-1 Mapping (IP + Port Number)
Public IP Efficiency	Low (Requires 1 public IP per active host)	Extremely High (Thousands of hosts share 1 public IP)
Scalability	Limited by size of allocated IP pool	Up to ~65,000 simultaneous connections per IP
Lab Outcome	PC3 failed due to pool exhaustion (2 IPs for 3 hosts)	All 3 PCs communicated simultaneously via 203.0.113.1